

A Diminished Reality Simulation for Driver-Car Interaction with Transparent Cockpits

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ABSTRACT

We anticipate advancements in mixed reality device technology which might benefit driver-car interaction scenarios and present a simulated diminished reality interface for car drivers. It runs in a custom driving simulation and allows drivers to perceive otherwise occluded objects of the environment through the car body. We expect to obtain insights that will be relevant to future real-world applications. We conducted a pre-study with participants performing a driving task with the prototype in a CAVE-like virtual environment. Users preferred large-sized see-through areas over small ones but had differing opinions on the level of transparency to use. In future work, we plan additional evaluations of the driving performance and will further extend the simulation.

Index Terms: H.5.1 [Information Interfaces and Presentation]: Multimedia Information Systems—Artificial, augmented, and virtual realities; H.5.1 [Information Interfaces and Presentation]: User Interfaces—User-centered design

1 INTRODUCTION

In recent years, novel user interface technologies in the automotive domain need to satisfy ever-increasing driver demands for handling driving-unrelated or tertiary tasks while retaining or improving safety. In the case of information output and visualization, much research has been focused on head-up display (HUD) interfaces. This includes the integration of augmented reality (AR) elements on the windshield. In comparison, the opportunities and implications of using head-mounted displays (HMDs) are still underexplored. We anticipate increased capabilities and convenience for see-through HMD technology and suggest a diminished reality (DR) interface allowing the driver to perceive occluded objects all around the vehicle. Thus, drivers could be warned about otherwise undetectable hazards or they might develop better distance estimation for low-height obstacles. Specific visual representations of occluded objects and additional visual hints could be used to direct the driver's attention and to convey critical information as fast as possible. In this paper, we suggest a simulation of such a see-through interface in a custom mixed reality (MR) driving simulator, focusing on user experience and interaction behavior rather than on current technical requirements for a real-world realization.

Existing research on see-through functionality in the automotive domain has mostly been focusing on diminishing other cars or obstacles to allow drivers to see further ahead. Yasuda and Ohama [2] examined see-through visualizations for walls at blind corners. They found simplified visualizations of occluded vehicles to be sufficient for distance estimation. Car manufacturers have already advertised with videos of simulated transparency effects for the car body, yet little research has been done on the interaction aspects of such systems. An actual real-world prototype has been

introduced by Tachi et al. [1] using a built-in projection system. Outside images are captured by mounted video cameras, then processed and projected within the car on special projection sheets that cover the car's interior. This shows that non-HMD see-through systems for drivers might be feasible as well, which further warrants examination of the user experience while driving transparent cars.

2 PROTOTYPE AND PRE-STUDY

The current prototype is a simulated DR interface in a custom MR driving simulation in Unity. It consists of a circular see-through spot (figure 1) which allows the driver to perceive parts of the surrounding environment otherwise occluded by the car. It is realized by using one additional virtual camera per eye with a specific culling mask to only render the respectively desired objects. Both cameras use off-center projection with a target projection plane that should be identical to the desired see-through spot in position, rotation and scale. Each camera renders to a texture which is then applied to that eye's instance of the target plane.

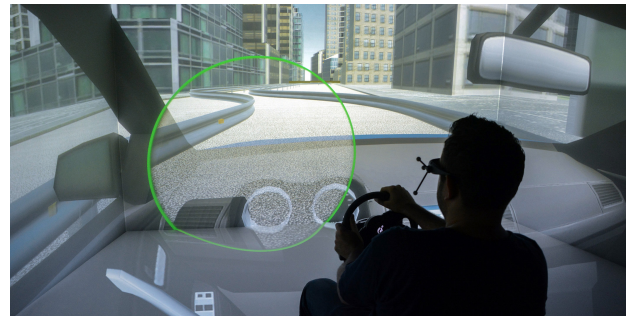


Figure 1: The driving simulation with enabled see-through effect running in a 4-sided CAVE-like virtual environment. Here, the see-through area is small-sized and using semi-transparency for the user car. For better visibility, rendering is 2D. Best viewed in color.

The behavior and appearance of the see-through effect can be adjusted with several properties. It can either align to the rotation of the driver's head or be placed at arbitrary places within the car, e.g. to draw attention to a specific hazard. The shape and degree of transparency can be varied by combining the render texture with an alpha mask shader. This also allows to regulate how hard or soft the blending at the margin should be. Car parts can be hidden individually and occluded objects can be shown with visual representations other than their usual exterior. For instance, one could provide simplified representations of obstacles that might require less visual or attentional demand to be comprehended.

In order to get first insights into the user experience and to find possible major shortcomings at an early stage, we conducted an informal pre-study with 8 participants (3 female) aged between 19 and 39 years ($M = 27.38$, $SD = 5.83$) and naïve to the experiment. It consisted of a driving task where participants had to steer a car through specific one-lane tracks delimited by guardrails under different conditions of the see-through system. This is intended to

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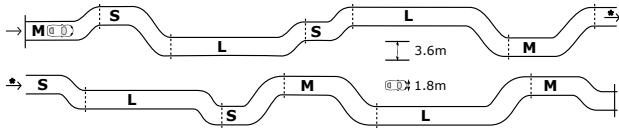


Figure 2: The base layout of the study track. The 500-meter track is made up of 12 segments, each consisting of a straight (S:short, M:medium or L:long) and a curve (left or right, across 1 or 2 lanes).

be a more demanding replication of a typical situation at highway road work sites when original lanes have to be diverted to narrow temporary lanes. The base layout (figure 2) was mirrored horizontally and vertically to reach a total of 4 usable tracks and constrain track learning effects. Participants were driving at a forced constant velocity of 40 kph (ca. 25 mph) and thus only had lateral control of the car plus control over the see-through area's position via head rotation. The within-subjects study was set up with a balanced Latin Square 2x2x2 factorial design with the independent variables see-through spot size (small: $d = 34^\circ$ and large: $d = 78^\circ$), see-through transparency (full and semi) and guardrails height (regular and ground). Figure 3 shows all relevant factor levels.

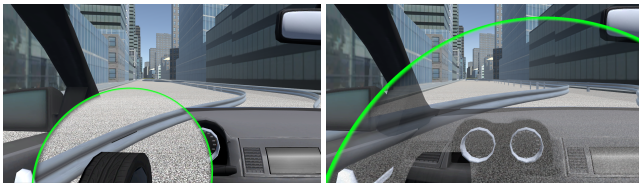


Figure 3: Screenshots of the see-through interface used in the study. Left: small, full-transparency see-through with ground guardrails. Right: large, semi-transparency see-through with regular guardrails.

We recorded user input (steering, head position and rotation), the user car position and direction and the lateral deviation from the ideal line (defined by the center between the delimiters). We plan to use this later in a more detailed analysis of the driving performance. We instructed participants to stay centered between the delimiters at all times. An actual collision with the guardrails was not possible, i.e. participants could recover without losing speed. In the beginning, a training phase allowed each participant to get used to the simulation. All factors and levels were explained during training. When the participant felt confident to start the experiment, the recorded driving task was started. It consisted of 12 runs in total: 8 runs with see-through in each possible condition and 4 baseline runs without (before regular runs 1, 3, 5 and 7). Thus, we were able to record the change in lateral deviation in relation to each participant's individual baseline skill. Multiple baseline runs further allow to limit learning effects influencing the final results.

To assess user experience and to receive additional qualitative feedback, we conducted a short post-test survey. Participants were asked to rate their comfort in using the system, its helpfulness for completing the driving task and potential irritation with the see-through spot's movements. All three aspects were rated (1-10) specifically for each of the prototype's different key features: small-sized, large-sized, semi-transparent and fully transparent.

3 RESULTS AND FUTURE WORK

The evaluation of the questionnaire results (figure 4) gives clear indications about the design of the see-through spot size. We applied a Wilcoxon signed-rank test for the difference of large-sized and small-sized spots for the ratings of comfort ($Z = 2.533$, $p = 0.011$, Cohen's $d = 0.896$), helpfulness ($Z = 2.533$, $p = 0.011$, Cohen's $d = 0.896$) and irritation ($Z = 2.370$, $p = 0.018$, Cohen's

$d = 0.836$). We can thus reject the Wilcoxon null hypothesis that there is no difference between the small-sized and large-sized see-through effect for all three items. The better ratings for large-sized see-through are also backed by additional feedback from the participants. Several users stated that the smaller version creates a higher attentional and visual demand. This is likely due to the movements induced by head-tracking which are perceived more severely in contrast to a larger see-through area. One participant noted that eye-tracking might lessen irritation with the moving small circle. Participants also generally preferred to be able to see both front tires at the same time, which was only possible with the larger version.

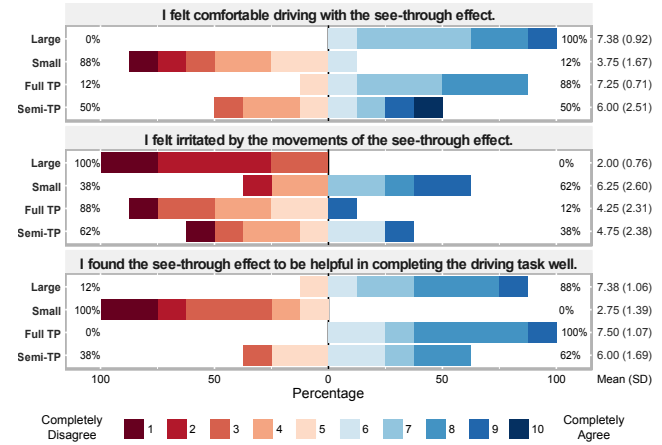


Figure 4: Results from the post-test questionnaire. Questions are targeting perceived comfort, helpfulness and irritation when using the see-through interface in its different conditions. Best viewed in color.

Regarding the differences between full and semi-transparency, participants were not as united. There is little difference regarding feelings of irritation between both conditions. Possible differences in comfort ($Z = 0.994$, $p = 0.320$) and helpfulness ($Z = 1.897$, $p = 0.058$) are more pronounced, but the null hypothesis of the Wilcoxon signed-rank test can not be rejected. The participants' feedback gives a similar impression: some criticized the semi-transparent version for drawing attention to the wrong things and it was perceived slightly less helpful in performing well for the driving task. Others however were unsatisfied with full transparency because it would feel more unreal and the total absence of the hood or front of the car might make them feel uneasy in a real-world scenario. One participant stated that the large-sized semi-transparent see-through system was "better than reality" because it would provide him with added value while not changing his visual patterns.

As part of our future work, we plan on evaluating the recorded driving task measurements to obtain objective results concerning potential differences in driving performance under the various see-through conditions. This will also include the question whether see-through capabilities support the handling of near-ground obstacles in particular. We plan to further extend the prototype by adding multiple visualization styles for occluded objects and we want to examine which representations most benefit the driver. To that end, we also want to explore more driving scenarios in which drivers might profit from see-through capabilities, e.g. parallel parking.

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